

Clipchamp: Setting the benchmark for simple, seamless video creation experience

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[About Clipchamp](#)

Clipchamp demo
video creation for
of users worldwide
creating a seamless
experience with the
integration of Google
Google Workspace
through its collaboration
with Chromebook



About Clipchamp

Clipchamp is a global video creation platform that gives anyone the tools to make high-quality videos, without relying on expensive equipment, complicated software, or previous video production experience. Founded in 2014, the Australian-based company has grown to a team of 80, serving more than 16 million registered users across 390,000+ businesses around the world.

Industries: Technology

Google Cloud results

- Easy out-of-the-box video editing capabilities with Chromebook
- Saves users 20 seconds per file upload with drag-and-drop capability supported by Chrome OS
- Delivers 99.99% availability
- Features scalability to support planned cloud-based video processing service

Support
than 1
video
world

Location: Australia

Datastore

Google Cloud (<https://cloud.google.com/>)

Logging (<https://cloud.google.com/logging/>),

App Engine (<https://cloud.google.com/appengine/>)

Cloud Pub/Sub

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Google Drive

Cloud Datastore

(<https://cloud.google.com/datastore/>)

Calendar

Logging (<https://cloud.google.com/logging/>)

Dataflow (<https://cloud.google.com/dataflow/>)

(<https://workspace.google.com/products/meet/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Gmail

Google Workspace

(<https://workspace.google.com/>)

Chromebook

Google Drive

(<https://workspace.google.com/products/drive/>)

Calendar

(<https://workspace.google.com/products/calendar/>)

Google Meet

(<https://workspace.google.com/products/meet/>)

Gmail

(<https://workspace.google.com/products/gmail/>)

Chromebook

(<https://chromeenterprise.google/devices/>)

The demand for video content and consumption is
on the rise

(<https://www.forbes.com/sites/falonfatemi/2021/02/01/how-the-pandemic-has-changed-video-content-and-consumption/?sh=3a1291ba6ec0>)

. Whether it's for personal use or for organizations to drive marketing and sales, more people are turning to video as a way to communicate. Seeing the need for a fuss-free video creation platform, Brisbane-based business Clipchamp (<https://clipchamp.com/en/>) aims to democratize access to video creation for businesses and individuals worldwide.

Founded in 2014, Clipchamp serves more than 16 million registered users worldwide. Its user-base spiked by 160% in 2020 when the pandemic hit and more people began to use video as a form of communication. "We see a diverse range of people using our platform. Whether it's for personal use like birthday videos for family members, or creating sales videos to reach out to customers, we cater to anyone who wants to make professional videos quickly," says Benjamin Gauthey, Vice-President of Global Strategy and Business Development at Clipchamp.

Clipchamp started operations in a cloud service. However, its client-centric model meant only a couple of static assets had to be served from the cloud. "We began looking for a more powerful, full-featured cloud service when we started creating user accounts," explains Alexander Dreiling, CEO and co-founder of Clipchamp. "We made a strategic decision to use a fully managed service so we could focus on our 'secret sauce'—enabling users to create videos in browser environments." Starting on a

hybrid model, Clipchamp today runs fully on Google Cloud (<https://cloud.google.com/>).

"Our mission has always been to make video really easy. We believe that video creation should be as easy as putting text on a piece of document, with little to no delay at any step of the user journey, and this is exactly what Google Cloud and Chromebook has enabled."

—Sören Balko, co-founder
and CTO, Clipchamp

Creating professional videos in minutes with Chromebook

In 2021, Clipchamp partnered with Chromebook (<https://www.google.com/chromebook/>) as the default video editing platform. Now, all Chromebook users can have access to Clipchamp and its video-editing features out of the box. "There's no installation or maintenance needed. You basically pop this thing open and get started on your project," says Sören Balko, co-founder and CTO of Clipchamp.

Unlike other operating systems that may require users to install software prior to use, the collaboration with Chromebook is a deliberate effort to remain ahead of its competitors, starting with a seamless user experience. "Our mission has always been to make video really easy. We believe that video creation should be as easy as putting text on a piece of document, with little to no delay at any step of the user journey, and this is exactly what Google Cloud and Chromebook has enabled," adds Balko.

Although there are hundreds of preset templates on the Clipchamp platform, the team realized the importance of allowing users to use their own footage. However, traditional cloud-based solutions would require massive waiting times when uploading large media files, resulting in a poor user experience. As a solution to this, Clipchamp runs fully in-browser, supported by the Chrome OS (<https://chromeenterprise.google/os/>), where the heavy lifting of video processing is done locally using hardware resources on the user's device.

Clipchamp also runs on WebAssembly, and with the support of [Chrome's WebAssembly SIMD](#)

(<https://v8.dev/features/simd>) (single instruction, multiple data), it has been able to speed up demanding workloads, including 4K video decoding and encoding.

With a drag and drop function, users can upload files onto the Clipchamp platform instantly, compared to the traditional waiting time of up to 20 seconds per file. While it may sound insignificant, this can easily add up with multiple files, not to mention disrupting the user experience.

Supporting growth with a fully managed service

Clipchamp is now supporting a user base of more than 15 million on [App Engine](#) (/appengine) and other Google Cloud products and is recording availability of more than 99.99%. And that user base keeps growing.

The business uses App Engine to run account creation, login-logout, and other management and billing functions, supported by a [BigQuery](#) (/bigquery) data warehouse and a [Datastore](#) (/datastore) NoSQL database service that stores project data, with [Cloud Storage](#) (/storage) storing media files users want to back up to Google Cloud.

“We do not get midnight phone calls about outages, we can scale to support continuous growth in user numbers, and we know skilled Google Cloud engineers are eliminating any security defects that

emerge in the technology stack,” says Dreiling. “So, with Google Cloud, rather than worry about backend stability, we can focus on ensuring our browser-based product continues to improve and provide an excellent experience for users.”

Dreiling described App Engine as easy to train new developers on, minimizing education costs and ensuring they can be productive quickly. “There is simply a group of endpoints, a database, and a technology stack on top of it—we can easily onboard new developers, train them, and let them get on with adding new features,” he says.

Balko adds that having the support of BigQuery has been a huge help in generating the data required to continuously improve its services. “What I find refreshing with BigQuery is how little you have to do to prepare a report. You basically write up the SQL prerequisites and that’s about it. The simplicity gives you this extra flexibility to basically just modify your query without worrying about the system.”

For data migration, Clipchamp uses Dataflow (/dataflow) for a fast, simplified streaming pipeline with low data latency. To monitor its backend performance in real time securely, the company relies on Cloud Logging (/logging).

While performance and scalability applies only to the cloud-based functions Clipchamp uses, rather than heavy-duty video processing, this is likely to change in the near future. “We are looking to supplement our client-side approach with a cloud-based fallback option for users of devices that do

not have much storage space or CPU processing capacity,” says Dreiling. “Google Cloud and App Engine give us the confidence to add this service without being concerned that the increased processing load will overwhelm our backend infrastructure.”

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—Sören Balko, co-founder
and CTO, Clipchamp

Simplifying the video editing process with Speech-to-Text

Clipchamp is now exploring Speech-to-Text (/speech-to-text) to help users create videos quickly and accurately.

"We found that Speech-to-Text gives really accurate transcriptions even with different accents. It also gives the timestamps, which is really useful because you can track down to the millisecond when a word starts or ends. That's really helpful when editing videos," says Balko.

The accuracy of Speech-to-Text means that video creators can simply click on a word uttered in a video, and they get taken straight to the exact timing in the video, down to the millisecond, where they can then choose to cut. This makes video editing a much quicker process since users don't have to manually move their cursor along the sound file to identify the exact moment where they want to trim the videos.

Another benefit of Speech-to-Text is its ability to support more than 125 languages and variants with accuracy, giving Clipchamp its global reach and appeal.

Enhancing team collaboration with Google Workspace

Internally, the Clipchamp team uses Google Workspace (<https://workspace.google.com/>) as its main collaboration hub. Employees use Gmail (<https://workspace.google.com/products/gmail/>) on a daily basis, organize and schedule events and meetings on Calendar (<https://workspace.google.com/products/calendar/>), and share documents and files on Drive (<https://workspace.google.com/products/drive/>). When meetings are held, the team gets together on Google Meet (<https://workspace.google.com/products/meet/>).

"We use other tools as extensions and integrations, but Google Workspace sits in the middle of what the team uses to collaborate and communicate. The managed Google accounts provisioned through Workspace also allow our team to easily log in to lots of third-party apps, doing away with having to use email/password, providing an extra level of convenience and security," says Balko.

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*—Alexander Dreiling, co-
founder and CEO,
Clipchamp*

Reaching more video creators around the world with Google Cloud

Clipchamp sees Google Cloud playing a key role in its continued expansion. “The internet today is moving towards video, and every business is catching on to that,” says Dreiling. “However, the problem they have is that video is typically expensive and time-consuming to produce. We want to be the platform for the 95% of businesses that have neither the large budgets nor the skills to understand how to create a very professional-looking video, and we see Google Cloud enabling us to do that.”

Future plans for Clipchamp include migrating its datastore instance to [Firestore](#) (/firestore), which offers a less rigidly constrained data model and better transaction throughput. The company is also looking to use [App Engine Flex](#) (/appengine/docs/flexible) and [Cloud Run](#) (/run) for more flexibility on new workloads like collaborative editing. "Essentially, we want a more comprehensive, fully managed back end and believe that App Engine Flex and Cloud Run will be good stepping stones, while giving us flexibility in resource allocation to customize our backend stack," says Balko.

"As the pandemic continues, consumers, educators, and businesses have realized the importance of video to bridge the communication gap, and Clipchamp can be used to create any video they may need," says Clipchamp CEO Alexander Dreiling. "Our continued partnership with Google enables us to reach even more remote video creators worldwide so they can quickly and easily create, edit, and export professional, meaningful, and targeted business videos."